# Project Definition Document

# Adaptable AI Camera Embedded Surveillance Drone

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# Contents:


1. **Project background and purpose**

    1.1) - Introduction [Page 3]
    1.2) - Objectives [Page 3]
    1.3) - Scope [Page 4]
    1.4) - Deliverables [Page 4]
    1.5) - Constraints [Page 4]
    1.6) - Assumptions [Page 4]

2. **Project rationale and operation**

    2.1) - Project benefits [Page 5]
    2.2) - Project operation [Page 5]
    2.3) - Options [Page 5]
    2.4) - Risk analysis [Page 6]
    2.5) - Resources [Page 7]
    2.6) - Ethical and legal considerations [Page 7]
    2.7) - Commercial considerations [Page 7]

3. **Project methodology and outcomes**

    3.1) - Initial project plan [Page 8]
    3.2) - Tasks and milestones [Page 8]
    3.3) - Schedule Gantt chart [Page 9]
    3.4) - Project control [Page 10]
    3.5) - Project evaluation [Page 10]

4. **References**

# 1. Project background and purpose

## 1.1. Introduction

This report outlines the initial conceptualisation and project planning considerations made in order to develop an adaptable AI camera embedded surveillance drone. The report is split into three sections. These sections include:

- Project background and purpose - Looks at the objectives, scope, delivery, constraints and assumptions made in the process of developing the project.
- Project rationale and operation - Explores the benefits, operation, options, risks and resources involved in the development of the project.
- Project methodology and outcomes. - Outlines the initial project plan, tasks and milestones of the project. This section will also include a project Gantt chart, project control procedure and project evaluation protocol.

By taking an in depth look at these sections the planning requirements for the development of an adaptable ai camera embedded surveillance drone can be analysed.

## 1.2. Objectives

The objective of this project is to develop a surveillance drone simulation through deep reinforcement learning (Schulman et al., 2017). This should allow the drone to observe a small area through an embedded camera for detecting and monitoring any suspicious activity as well as adapting to any changes in the environment. The drone simulation within the simulation suite has a camera module that can be used to replicate a real camera from a surveillance drone that utilises machine learning and AI to detect and classify visual data as suspicious or not. The drone simulation will be able to make flight control decisions based on the incoming visual data from the camera module. These decisions may include:

- capturing a snapshot of suspicious activity/objects.
- tracking any suspicious objects/activity detected.
- travelling within range of a designated target or area of activity where clearer image data can be captured.
- A detected suspicious target can be followed.

For practicality and simplicity the integration of computer vision for detection of suspicious activity and simulation of the flight based decision making will be the primary objective of this project. This will also be used to assess the success and effectiveness of the project when conducting final evaluations and observations.

### 1.3. Scope

This project will consist of a camera embedded drone simulation that can detect and classify suspicious activity from visual data using computer vision and a flight algorithm that takes the visual data as input and outputs a set of flight control instructions to effectively make surveillance based decisions that change depending on environmental factors. This project will not include a physical drone or camera as one can be modelled using simulation software. A camera module in the simulation suite will be used for visual data capture and detection of suspicious activity/objects, however pretrained data and computer vision models will be utilised for practicality and simplification of the overall project.

### 1.4. Deliverables

The project will be delivered as a simulation using Webots simulation suite for the drone flight algorithm. This will be accomplished using a  python controller script. Additionally, a camera plugin, will be used for displaying a live feed in the simulation suite that can be used for object detection and tracking with open CV (Cyberbotics, n.d.). The finalised project will be capable of making decisions regarding its flight based on input data gathered from the camera. The necessary hardware will include a system that is capable of running Webots simulation suite or any other simulation software that can effectively simulate the objectives of this project.

### 1.5. Constraints

Commercial surveillance drones use relatively expensive cameras to capture data with much higher clarity which allows for more robust and accurate detection and classification vision models to be implemented. As such a major constraint on the success of this project relies on the ability to detect and classify with the use of a lower resolution camera that is more easily accessible. Another constraint is the accessibility and modelling capabilities of the simulation software chosen to develop the project, as well as the size of the data set and the computer vision detection and classification model that will be chosen for implementation to detect suspicious activity. Ultimately these constraints can be mitigated with more in depth research into the capabilities of multiple drone simulation software that is accessible online.

### 1.6. Assumptions

The assumptions being made through the development of this project include the assumption that the desired simulation software is capable of simulating both the necessary flight dynamics of a surveillance drone and also the assumption that the camera plugin in the simulation suite is compatible with detection and classification computer vision models. Another assumption that is made is that training data will be easily accessible for training the computer vision model for specific detection and classification tasks. These assumptions are justified as there is no guarantee that without the physical hardware the simulation for an adaptable AI camera embedded surveillance drone will behave as intended given model and controller constraints that may be presented by the chosen simulation software.

## 2. Project rationale and operation

### 2.1. Project benefits

There are various benefits gained from the development of surveillance drones adapted with AI cameras. These benefits include:

- Experience - A better understanding of a growing technology that is widely used in agriculture, construction, geology, archeology, search, rescue, military and many more applications around the world.
- Enhanced security - surveillance drones are able to investigate and observe hazardous and potentially dangerous areas where operators may be at risk. This allows for improved security to oneself and others, including security personnel, law enforcement, and rapid response or search and rescue teams.
- Real time data and tracking - drones allow for real time tracking and data through various sensors and cameras that transmit a live feed to a ground station or flight control software.
- Better cost efficiency - Drones cost much less to operate and maintain compared to extensive security camera networks or on site security services.
- Wide area coverage - Drones are able to observe a wider area of coverage compared to traditional security cameras, as well as reaching areas security cameras and security personnel cannot reach.

### 2.2. Project operation

This project will be operated Using Webots simulation software for the development and deployment of a drone model with flight control (PX4 Autopilot, n.d), an additional detection model and script will be used in accordance with a camera plugin available in the Webots simulation suite as a drone model attachment. The success of the project will be assessed on the detection of suspicious activity and the drone models response to it.

### 2.3. Options

Alternative options to Webots available for simulating the project include ROS2, Unreal Engine, Unity, Matlab and Simulink. However, these alternative options may be more difficult to set up for this specific task. Furthermore, these alternative methods might not support all three components of flight dynamics, 3D visual environments and computer vision integration.

### 2.4. Risk analysis and mitigation

| Risk | Risk type | Potential effects | Impact | Likelihood | Mitigation |
|---|---|---|---|---|---|

| | | of risk | of risk (1 - 5) | of risk occurring (1 - 3) | |
|---|---|---|---|---|---|
| Hardware failure | Technical | Inability to complete project development due to lack of development resources. | 5 | 1 | Hardware can be maintained and work can be saved regularly to ensure the health and functionality of the hardware as well as ensure there are back ups of work in case any problems occur. |
| Delays to development | Project management | Delays to development could result in the project not being delivered on time or successfully. | 3 | 2 | Strict planning and development schedules can be employed to reduce the likelihood of delays in development due to insufficient planning. |
| Lack of training data | Project management | Lack of data could result in suboptimal training and poor performance. | 2 | 3 | By exploring various dataset providers such as kaggle and setting aside enough time for development to gather my own data if necessary, it is possible to reduce the risk of a lack of training data. |
| Changes to the development platform | External | The platform changing could lead to fundamental design changes. | 4 | 1 | By looking at possible alternatives to the development platform chosen, it will be possible to mitigate the risk of any major changes to the development platform that may impact the project. |
| Scope creep | Project management | The scope of the project creeping could result in wasted time due to prolonging. | 1 | 1 | By outlining a clear scope and set of objectives in planning documentation, and adhering to them it will be possible to reference this and reduce the likelihood of scope creep. |
| Time overrun | Project management | Time overrun leads to delays in project delivery. | 2 | 2 | Strictly following planning documentation such as a gantt chart and project timeline can reduce the likelihood of tim overrun. |
| Prolonged Illness | External | Prolonged illness can lead to schedule and planning delays which directly impact development. | 4 | 1 | Prolonged illness can be mitigated by setting aside enough development time to account for any delays from illness as well as setting aside enough time to recover if possible. This will ensure should illness occur, it is within expectation and does not interfere with the timely completion of the project. |

### 2.5. Resources required

The only other resources required other than the standard software and hardware being used to run the software is a good resolution camera module or webcam which is compatible with the Webots simulation suite. Another important resource which will be required is a surveillance drone image and video dataset and an Open CV detection and classification model.

### 2.6. Ethical and legal considerations

Various ethical and legal considerations were made for the development of this project in order to ensure fairness and legal commitments are met along the development process (UK Parliament, 2018) .

The Ethical considerations made include:

- Privacy and data collection
- Bias and discrimination
- Accountability
- Proliferation and misuse

The legal considerations include:

- Privacy and data protection laws
- Airspace and aviation regulations
- Software licensing and intellectual property
- Liability and product safety

### 2.7. Commercial considerations

The primary commercial considerations made regarding this project include:

- Marketability of the project - the market for drone technology is already vast, even more so for surveillance and security drones which have become much more valuable and costly given the increased risk to global security over the past decade resulting in high demand among security services, law enforcement, military and enthusiasts.
- Cost of delivering the project - As the project will be software based, the development costs are especially low.
- Competitors - Vizgard, Teledyne FLIR and Palladyne AI are some of the competitors supplying demand for specialised edge AI software similar to what I will be developing.

Other commercial considerations explored for this particular project include:

- Monetisation of the software - As Webots is an open source software it can be used to develop commercial software without the need for any royalties being paid to the platform.

- Market compliance and barriers to entry - U AI systems used for surveillance are heavily regulated under the EU AI act, especially systems monitoring public areas. The prerequisites for meeting regulations such as third party conformity assessments and risk management systems may be costly to set up before any software can be monetised legally (European Parliament and Council of the European Union, 2024).

## 3. Project methodology and outcomes

### 3.1. Initial project plan

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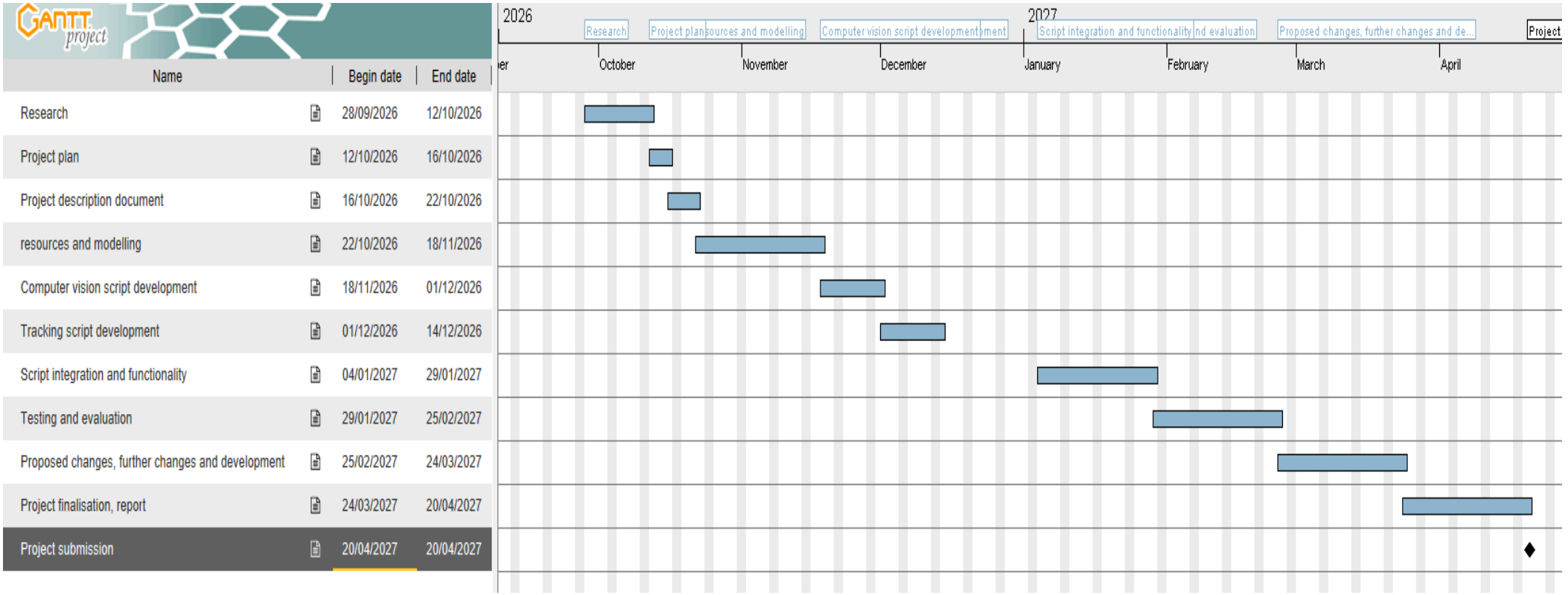
**Research → Documentation → Resource gathering → Development → Testing → Review**

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### 3.2. Tasks and milestones

1. Project research.
2. Project planning documentation.
3. Project description document.
4. Download relevant simulation software (Webots simulation suite).
5. Download Ardupilot firmware or PX4 Autopilot source code.and configure in Webots.
6. Design or utilise premade drone models in the Webots simulation suite.
7. Develop companion computer code in Webots.
8. Write a Webots controller script that retrieves camera feed and sends it to a tracking script.
9. Write a tracking script that just processes image arrays and calculates target coordinates which can be sent as movement commands.
10. Conduct rigorous testing and refine the system to improve accuracy and repeatability.
11. Conduct an in depth evaluation of the projects development and completion.

### 3.3. Gantt chart


GANTT project
Name
Begin date
End date
Research 28/09/2026 12/10/2026
Project plan 12/10/2026 16/10/2026
Project description document 16/10/2026 22/10/2026
resources and modelling 22/10/2026 18/11/2026
Computer vision script development 18/11/2026 01/12/2026
Tracking script development 01/12/2026 14/12/2026
Script integration and functionality 04/01/2027 29/01/2027
Testing and evaluation 29/01/2027 25/02/2027
Proposed changes, further changes and development 25/02/2027 24/03/2027
Project finalisation, report 24/03/2027 20/04/2027
Project submission 20/04/2027 20/04/2027
2026
2027
October
November
December
January
February
March
April

### 3.4. Project control

Project control will be carried out on a weekly basis using a progress checker which can be used to monitor the development of the project at various key stages.The progress checker is delivered as a table which can be populated weekly to track and update progress.

### 3.5. Project evaluation

A number of methods for evaluation will be used to determine the success and performance of the project. These methods include:

- User evaluation - The functionality of the software can be tested in the Webots simulation suite and evaluated by users to identify any changes for improvement.
- Self reflection - self reflection and review can be carried out at each project milestone and recorded using documentation such as a project report to, evaluate and justify any design considerations or changes made to improve the success of each project task.
- Evaluation metrics - Mathematical and statistical metrics such as intersection over union to check bounding box accuracy, Cross track error for tracking the difference between the drones actual virtual position and its ideal position for centering the camera target. This data is  recorded by the software during automated runs and can be analysed to evaluate the success of the project.
- Automated batch testing - An automated testing pipeline can be used in Webots to evaluate the project in different scenarios and under different environmental conditions.

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## 4. References


Cyberbotics (n.d.) *Webots Reference Manual: Camera*. Available at: https://www.cyberbotics.com/doc/reference/camera (Accessed: 12 October 2026).

PX4 Autopilot (n.d.) *PX4 User Guide*. Available at: https://docs.px4.io/ (Accessed: 12 October 2026).

Schulman, J., Wolski, F., Dhariwal, P., Radford, A. and Klimov, O. (2017) *Proximal Policy Optimization Algorithms*. arXiv:1707.06347. Available at: https://arxiv.org/abs/1707.06347 (Accessed: 12 October 2026).

UK Parliament (2018) *Data Protection Act 2018*. Available at: https://www.legislation.gov.uk/ukpga/2018/12/contents (Accessed: 20 October 2026).

European Parliament and Council of the European Union (2024) Regulation (EU) 2024/1689 (Artificial Intelligence Act). Available at: https://eur-lex.europa.eu/eli/reg/2024/1689/oj (Accessed: 20 October 2026).